

WaterML2.0

Ispra workshop

Water for a Healthy Country

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National Research
FLAGSHIPS



Introductions

Intention

- Creation of an open, re-usable and *useful*, standard for the exchange of hydrological data sets



Topics

Planning

- Use cases & requirements
- Scope of first specification
- Roadmap and methodology
 - Creation of a work plan?

Technical

- Modeling Overview:
 - Overview of using EA with HollowWorld & FullMoon
 - Walk through of current WaterML2.0 model
- Observations & Measurements overview
- Grouping of observations
- Time series model
- Feature & sampling descriptions
- Procedural (processing, sensors etc.) information
- How to support code lists and inclusion for specific profile use?
- Development of SWE common profile

Scope

Observation style	Description
In-situ, fixed observation style	Generally temporally dense, spatially sparse, small number of observed phenomena. Examples: river level or stage, river discharge, storage level, rainfall etc.
Ex-situ, complex processing observations	Temporally sparse, spatially sparse, many observed phenomena. Examples: nutrients (nitrate, phosphorus etc.), pesticides (atrazine, glyphosate etc.), biologicals, pH, turbidity etc.
Complex data products	These consist of processed or synthesised observational data, mainly created to provide estimation of not directly measurable phenomena or predictions of future values. Examples: outputs from models or algorithms, water storage estimates, calculation of complex physics-chemistry, biological indices (French : IBGN, German/Austrian : Saprobic Indice, ...).

Use cases (general categories)

- Water agency to a central repository
 - Community science, various 'data suppliers'
 - WDTF et al..
- Agency to agency transfer
 - Cross international/state borders (GW IE)
 - Cross federal agencies (NWS, USGS, COE – surface water...IE?)
- Commercial data vendor to consumer
- Central repository to external users (public etc.)
 - Different levels of public
 - Services stack
 - Also being used within the GW IE
 - How a web client/tool uses a service interface?

- Purpose of the transfer
 - Incremental exchange
 - System to system, system to archive
- Metadata or data transfer
 - Submitting for archival purposes
- Query mechanism for accessing observations data
 - What are the restrictions of waterml2
 - Query features starting with observations, or query observations starting with features... query based on phenomenon
 - How to query forecasts?

- Fitness for use of data:

- How to categorise the fitness of data for particular data?
 - USGS provide categories: provisional data,
 - Period of applicability/useability of a time series
- Encoding uncertainty in the data
- Ability to describe the various aspects of quality in more detail than simply a code...

- Process description

- Keep it lightweight
- Should we re-use an existing standard?
- MOLES for metadata, usage in UK data centres (Brian Lawrence)

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Thank you

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